

Telecom Customer Intelligence & Churn Prediction

Executive Project Report | EDA • BI Dashboard • Predictive Modeling

14.7%	43.6%	98.1%	98.6%	92.7%
Overall churn	International-plan churn	Model accuracy	ROC-AUC	Model F1

1. Outlook

Objective. Turn a customer-level telecom dataset into an actionable churn-intelligence workflow: identify churn patterns, communicate them through a Power BI dashboard, and predict customers at elevated churn risk using an optimized XGBoost gradient-boosted tree ensemble.

Manager value. The project connects descriptive analytics to predictive decision support. The output can help prioritize retention analysis, customer-service interventions, and targeted offers rather than treating all customers equally.

2. Overview & Introduction

The project analyzes 2,666 customer records with 20 features covering account characteristics, subscription plans, voice/data usage proxies, charges, international activity, and customer-service interactions. The workflow follows a practical analytics lifecycle: data validation → exploratory data analysis → business-oriented segmentation → Power BI visualization → churn classification → evaluation and interpretation.

3. Problem Definition

Customer churn is a business-risk problem because retention decisions require identifying which customer profiles are more likely to leave. The analytical problem was therefore defined as a binary classification task: **predict whether a customer will churn (True/False)**, while using EDA and BI to explain the strongest observable patterns behind churn. Because the dataset is a static snapshot, the findings should be interpreted as associations, not causal effects or a measured time-to-churn forecast.

4. Key EDA Findings

- Overall churn rate is **14.7%**, meaning churn is a minority class and the retention population is substantially larger.
- International-plan customers show a **43.6%** churn rate versus **11.2%** for customers without the plan—one of the clearest segmentation signals in the analysis.
- Churned customers average about 205.2 day-time minutes versus 175.1 for retained customers, indicating materially higher daytime usage among churners.
- Voice-mail-plan adoption is lower among churners: approximately **8.9%** versus **16.7%** among retained customers.
- Customer-service interaction is strongly associated with churn at higher call counts: churn reaches about **48.1%** at **4** service calls and **40.8%** at **5** calls. Very high call-count categories have small samples and should not be overinterpreted.
- International usage is only modestly different between churn groups, so international minutes alone are a weaker signal than the international-plan segment.
- Corresponding charge variables closely track their usage-minute variables; they provide limited independent information and should be interpreted with awareness of redundancy/multicollinearity.

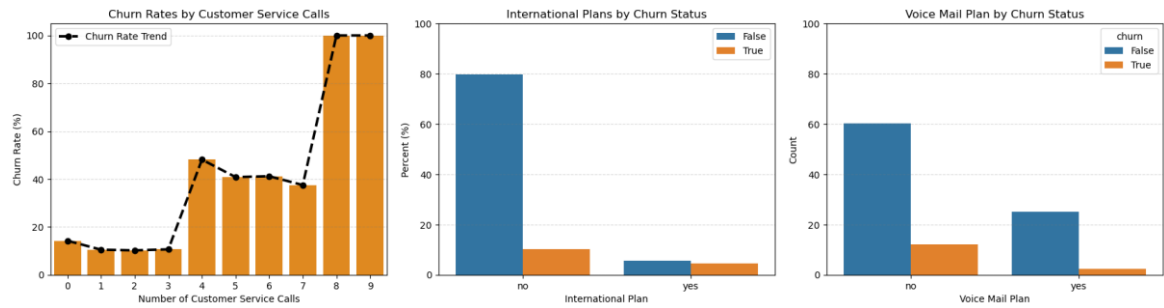


Figure 1.1 – Customer Churn by Different Parameters.

5. Business Intelligence Dashboard

The Power BI dashboard translates the EDA into an executive monitoring layer. It provides a compact view of overall churn, segment comparisons, and behavioral/service patterns so that stakeholders can move from “how much churn?” to “which customer groups look different?”

Dashboard view	Decision question
Churn rate by international plan	Which subscription segment has disproportionate churn?
Churn rate by voice-mail plan	Does plan adoption distinguish retention patterns?
Churn rate by customer-service calls	Where does support interaction become a potential churn-risk signal?
Usage / customer profile views	Which usage and account characteristics differ between churn groups?

6. Predictive Modeling

A supervised binary classification model was trained to predict churn. The selected approach is XGBoost, a gradient-boosted decision-tree ensemble that builds trees sequentially to correct previous model errors. The model was optimized for the project and evaluated on a held-out test set.

98.1% Accuracy	98.8% Precision	87.4% Recall	92.7% F1-score	98.6% ROC-AUC
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Interpretation. The 98.6% ROC-AUC indicates strong ranking/discrimination capability on the test set. Precision of 98.8% means predicted churners were highly reliable, while recall of 87.4% indicates that the model captured most—but not all—actual churners. F1-score of 92.7% provides a useful balance between these two objectives. Accuracy is reported for completeness, but should not be the primary business metric given the imbalanced churn distribution.

7. End-to-End Workflow

DATA → EDA → BUSINESS QUESTIONS → POWER BI → FEATURE ENGINEERING → XGBOOST → TEST EVALUATION → ACTIONABLE RISK SIGNALS

8. Future Work

- Add time-based customer history to move from static churn classification toward early-warning churn prediction and a defined prediction horizon.
- Use probability calibration and a business-selected decision threshold instead of relying only on the default classification threshold.
- Evaluate PR-AUC, confusion matrix, lift/gain and cost-sensitive metrics to quantify the operational value of contacting predicted churners.
- Apply model explainability (e.g., SHAP) to identify the strongest customer-level drivers and make model outputs easier for business teams to act on.
- Validate the model on a temporally newer or external dataset to test robustness and reduce the risk of overestimating offline performance.
- Connect dashboard segments to retention actions and measure intervention outcomes, creating a closed-loop churn-management process.

9. Executive Conclusion

This project demonstrates an end-to-end customer analytics capability: it identifies meaningful churn segments through EDA, turns those findings into a decision-oriented Power BI dashboard, and builds a high-performing XGBoost classifier for customer-risk prediction. The strongest observable signals include international-plan membership, higher daytime usage, voice-mail-plan adoption, and repeated customer-service interactions. The next maturity step is not simply a more complex model; it is operational validation, explainability, threshold optimization, and linking predictions to measurable retention actions.

Method note: Percentages are rounded. EDA relationships are associative; they do not establish causation. Test metrics reflect the stated held-out test set.